```
Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}
public void area(){
System.out.println("Area of Cylinder: "+(2*PI*radius*(radius+height)));
}
public void volume(){
System.out.println("Volume of Cylinder: "+(PI*radius*radius*height));
}
}
public class OperationDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Cylinder cy=new Cylinder(5,10);
c.area();
c.volume();
cy.area();
cy.volume();
}
}
```

Slip 22

1) Write a Java program to accept a number from user and print all prime numbers upto that number (Use Buffered Reader class).

```
import java.io.*;
public class PrimeNumbers{
public static void main(String[] args)throws IOException{
```

```
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
System.out.print("Enter a number: ");
int n=Integer.parseInt(br.readLine());
System.out.println("Prime numbers:");
for(int i=2;i<=n;i++){
int count=0;
for(int j=1;j<=i;j++){
if(i%j==0)
count++;
}
if(count==2)
System.out.print(i+" ");
}
}
}
```

2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;
public class StudentNames{
public static void main(String[] args){
ArrayList<String> list=new ArrayList<>();
for(String name:args)
list.add(name);
System.out.println("Using Iterator:");
Iterator<String> i=list.iterator();
while(i.hasNext())
System.out.println(i.next());
```

```
System.out.println("Using ListIterator:");
ListIterator<String> li=list.listIterator();
while(li.hasNext())
System.out.println(li.next());
}
}
```

OR

2)) Write a Java program to create LinkedList of String objects and perform the following:

i. Add element at the end of the list

ii. Delete first element of the list

iii. Display the contents of list in reverse order

```
import java.util.*;
public class LinkedListDemo{
public static void main(String[] args){
LinkedList<String> list=new LinkedList<>();
list.add("Apple");
list.add("Banana");
list.add("Mango");
list.addLast("Orange");
System.out.println("After adding at end: "+list);
list.removeFirst();
System.out.println("After deleting first element: "+list);
System.out.println("List in reverse order:");
Iterator<String> i=list.descendingIterator();
while(i.hasNext())
System.out.println(i.next());
```